

Alpha Adventist Digital School Platform — Audit & Master Blueprint

Sep 28, 2026 · @Humands

Executive summary

Alpha Adventist Pre & Primary School should move from a single brochure website to one governed **Alpha Adventist Digital School Platform (ADSP)**: public website, CMS, staff portal, HR, documents, workflows, admissions, parent and student portals, and a mobile app.

This document combines two inputs:

- **Part A — Website audit (28 September 2026).** A code-level review of the current site at alpha-adventist-pre-primary-school.onrender.com. It is well built, but close to invisible: it points search engines to a domain that does not exist, has no Kiswahili, and has security and child-safeguarding gaps to close first.
- **Parts B–H — Platform master blueprint.** The requirements for the full platform: users, architecture, modules, permissions, workflows, privacy, security, data model and phased delivery.

The order of work is fixed by risk: **stabilise and secure the current site first (P0)**, then grow visibility and the CMS, then the staff and HR modules, then admissions and the parent and student portals, and only then native mobile apps. The target outcome in one line each:

Who	Journey
Parent	Discover → Understand → Trust → Visit → Apply
Staff member	Login → Read notices → Access documents → Apply for leave → Track requests → View contract
Manager	Review → Approve → Monitor → Report
HR	Manage staff → Contracts → Leave → Documents → Compliance
CMS team	Create → Review → Approve → Publish → Archive
ICT team	Operate → Secure → Monitor → Back up → Recover
Executive management	Govern → Authorise → Monitor → Decide

Part A — Website audit findings

The current site is technically strong but scores 2/10 on visibility, because its canonical, sitemap and share links point to a domain that does not resolve. The review covered the public GitHub source and a local run of every page; items marked *verify* could not be checked on the live server.

A1. Scorecard

Area	Score /10	Reason
Code and security	7	Clean zero-dependency build; default-password and rate-limit weaknesses
SEO and visibility	2	Canonical, sitemap, share image and structured data point to alphaadventist.ac.tz, which does not resolve
Content and trust	4	27 "pending / to be announced" phrases on the Parents page; 13 on Computer Learning
Local fit (Tanzania)	3	English only; no Google Maps pin; Yahoo email address
Child safeguarding	6	Good privacy notice; no consent capture on forms or photos
Admissions conversion	4	Forms save to a file; nobody is alerted

A2. Critical fixes (P0 — this week)

1. **Dead domain in search and share tags.** `src/layout.js` hardcodes `https://alphaadventist.ac.tz` in the canonical URL, `og:url`, `og:image` and `School` structured data; `render.yaml` sets `BASE_URL` to the same domain. Use `BASE_URL` everywhere and set it to the live address until a real domain exists.
2. **Check the live robots.txt (verify).** The repository allows all crawlers, but an automated reader was refused by the live site. If the live file says `Disallow: /`, Google cannot list the school.
3. **Default admin password may be live (verify).** The README publishes `admin / Alpha@2026!`; `server.js` falls back to it when `ADMIN_PASSWORD` is unset; `data/db.json` holds that account's hash. Test the login, change it, and remove it from the README.
4. **Admissions enquiries go nowhere.** All five public forms only write to `data/db.json`. Add an email/SMS/WhatsApp alert per submission and give the parent a reference number.
5. **Public repository exposes child photos and internal files.** `uploads/` (11 raw WhatsApp photos, many of children), `work/` (14 MB of screenshots and a cookie file), `pdf_text.txt` (internal brief) and `.sudo_as_admin_successful`. Make the repo private or purge these from git history.
6. **Harden login.** Rate limiting trusts the spoofable `X-Forwarded-For` header and resets entirely at 5,000 entries. Add per-username lockout and a strong-password rule.

A3. Visibility levers, ranked

1. **Google Business Profile** with the exact Msimba pin, photos, hours and phones — parents search Maps first.
2. **Kiswahili version** of at least Home, Admissions and Contact, with an EN/SW switch. Target searches such as *shule ya awali na msingi Kigoma*, *shule za bweni Kigoma*, *Shule ya Kiadventista Kigoma*.
3. **Official domain and email.** `.sc.tz` usually fits a primary school (`.ac.tz` is generally higher education) — confirm with tzNIC; then `info@` and `admissions@` on that domain.
4. **Adventist backlinks** from the Western Tanzania Conference, the East-Central Africa Division and the Seventh-day Adventist Encyclopedia entry on the school.
5. **Results page** for examination centre PS0603108, updated yearly.
6. **Google Search Console** plus privacy-friendly analytics.

A4. Content and user-experience gaps

- Hide any card without real content; show three real things rather than nine empty boxes.
- Cut the homepage from 17 sections (about 2,100 words) to about 7: Hero, Why Alpha, Levels, Life at Alpha, News, Admissions, Contact.
- Add trust builders in this order: Head of School photo, fee range, term calendar, uniform and requirements list, boarding photos, teacher count and qualifications, 2–3 consented parent testimonials.
- Replace the 10 recycled photos with an organised photo day.
- Verify and publish the official social media accounts.
- Expand the Kids Zone with Kiswahili reading and weekly teacher-assigned quizzes.

A5. Child protection and law (confirm with a Tanzanian lawyer)

- **Personal Data Protection Act, 2022:** check registration duties with the Personal Data Protection Commission (PDPC); add consent checkboxes, a retention period and scheduled deletion for form data.
- **Photo consent:** signed parental media consent per child, renewed yearly; remove a photo within 48 hours on request; never name a child beside a photo.
- **Law of the Child Act, 2009:** publish a named safeguarding contact and a way to report concerns.

- **Kids Zone:** no external links, ads, chat or data collection from children.

A6. Documentation

Split the README into a public setup guide with no credentials and a private Operations Manual for the school office: posting news, answering submissions, adding photos, backups and a monthly checklist.

A7. Business model

- **Community Computer Training Centre:** one paid evening/weekend course first, with online registration and M-Pesa/Tigo Pesa payment instructions.
- **Holiday tuition and ICT camps** during breaks.
- **Sponsor-a-Child and alumni giving** aimed at the Adventist diaspora and partner churches, with an accountable donation page.
- **Admissions funnel metric** each term: enquiries → visits → enrolments.

A8. Social media

- Facebook Page and WhatsApp Channel first; TikTok and YouTube Shorts for the choir.
- Weekly rhythm: Monday learning moment, Wednesday faith, Friday achievement or ICT, Saturday (Sabbath) verse card.
- Bilingual posts that link back to a news page; every post follows the media-consent register.

A9. Mobile recommendation

Do not build a native app before the portals exist. First make the site installable (PWA: manifest exists, add a service worker), use WhatsApp Business and bulk SMS for notices, and build an Android-first app in the mobile phase.

Part B — Platform vision, users and architecture

The project is one digital ecosystem with access controlled by each user's responsibilities, not a website with many pages.

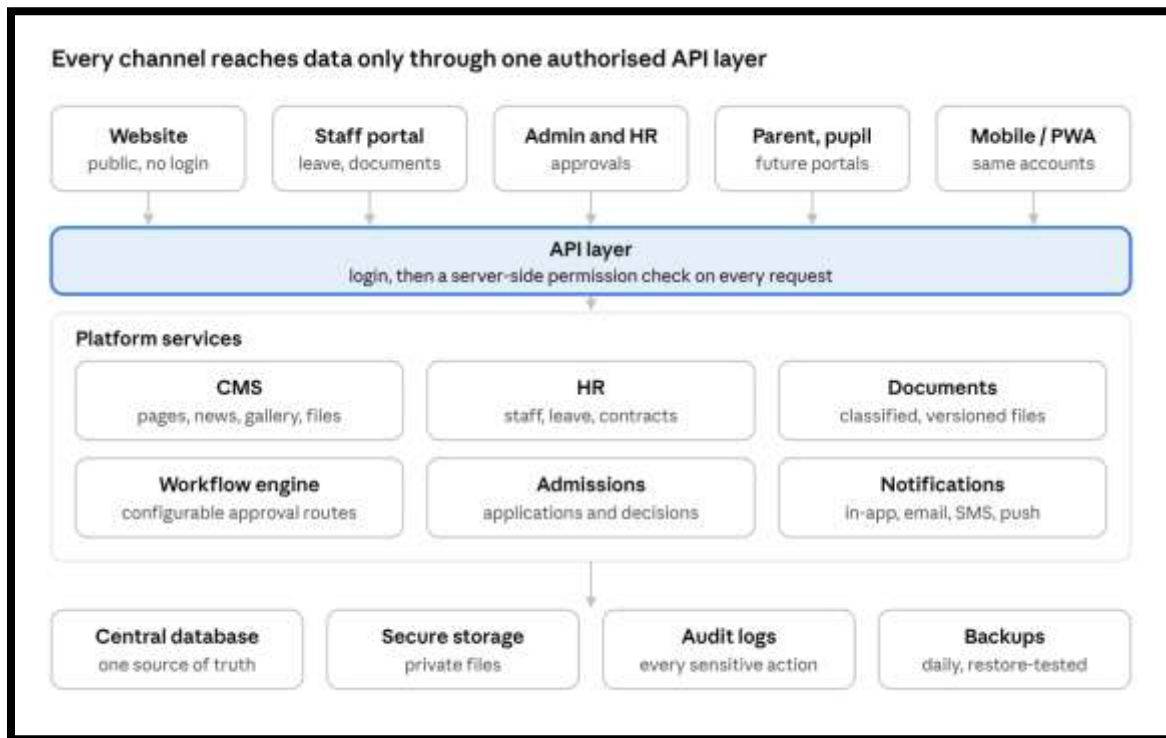
B1. Platform components

1. Public School Website
2. CMS — Website Controller
3. Staff Portal
4. HR Management
5. Leave Management
6. Staff Contract Management
7. Document Management System
8. Public Download Centre
9. Internal Staff Document Centre
10. Digital Staff Notice Board
11. Workflow and Approval Engine
12. Administration Portal
13. Admissions Management
14. Media and Gallery Management
15. Privacy and Consent Management
16. Reporting and Analytics
17. Notification Centre
18. Parent Portal
19. Student Portal
20. Progressive Web Application
21. Future Android/iOS Application

B2. Primary users

User	Access
Public users	Prospective and current parents, visitors, community, job seekers — public pages only
Students	Future authenticated academic services
Parents/guardians	Future services for the children they are authorised to represent
Teachers	Teaching resources, notices, own documents, leave, staff services
Supporting staff	Authorised HR and institutional resources
Heads of Department	Review departmental requests and documents
Academic Coordinator	Authorised academic information and workflows
HR Officer	Employee records, contracts, leave, HR documents
Finance Officer	Approved finance workflows only
Administration	Administrative processes and approvals
Head Teacher	Executive oversight and final approvals
ICT/System Administrator	Technical administration, without automatic access to confidential HR or business data
CMS/Communications Officer	Website content
DPO/Privacy Officer	Privacy and compliance oversight
Auditor	Read-only access to specifically authorised records

B3. Architecture



ADSP architecture · 5 channels, 1 API layer, 6 services, 4 data stores

Public pages need no login; every portal request passes the same API layer, which checks the user's permission on the server before any service touches data.

Part C — Public website and CMS

The public website exists to build trust and turn interested parents into visitors and applicants; the CMS lets authorised staff change it without a programmer.

C1. Navigation

Home | About | Academics | Admissions | Adventist Education | Student Life | Talents | ICT | News & Events | Gallery | Downloads | Contact | Portal Login

On mobile, keep a bottom bar with WhatsApp, Call and Apply (already built on the current site).

C2. Homepage

- **Title:** Alpha Adventist Pre & Primary School
- **Strapline:** Quality Education • Adventist Values • Technology • Talent Development
- **Supporting message:** Nurturing children spiritually, academically, intellectually, physically, socially and creatively for a purposeful future.
- **Primary actions:** Apply for Admission · Book a School Visit · Contact the School
- **Brand line:** Learn • Believe • Create • Discover • Grow

Per the audit (A4), the homepage should carry about seven sections, not seventeen, and a Kiswahili version.

C3. Why choose Alpha — six pillars

#	Pillar	Promise
01	Quality Education	Strong academic foundations and effective teaching
02	Adventist Christian Values	Faith, integrity, discipline, responsibility, service and good character
03	ICT and Digital Skills	Preparing pupils for the digital future
04	Talent Development	Discovering and developing each child's abilities
05	Safe Learning Environment	Safeguarding and child-friendly education
06	Whole-Child Development	Spiritual, intellectual, physical, emotional, social and creative growth

C4. Adventist identity

State clearly that Alpha provides education based on Seventh-day Adventist Christian values, within the Western Tanzania Conference – Kigoma. Core values: faith in God, integrity, discipline, respect, responsibility, compassion, service, healthy living, leadership, academic excellence, good citizenship and stewardship. This is Alpha's main differentiator.

C5. Discover Alpha Talents

Categories: choir, singing, musical instruments, sports, art, public speaking, debate, ICT, computing, science projects, creativity, leadership, cultural activities and academic competitions. Embed selected posts from the school's official YouTube, Instagram and Facebook accounts as evidence of real school life, not just footer icons — only content that has passed the media-consent check (F4).

C6. Download Centre

A public Downloads page, grouped by category:

- **Admissions:** admission form, requirements, prospectus, application instructions
- **Parents:** parent handbook, school rules, uniform guidance, school calendar
- **Academic:** public academic calendar and learning information
- **Careers:** job application forms, vacancy documents
- **Public policies:** only those approved for public access

Each file shows document, category, version, date, size and a download action; the system records version and publication date automatically. Public and private documents live in separate repositories: a confidential file must never become reachable because someone guesses its URL.

C7. CMS capabilities

Authorised users can create and edit pages, news and events; upload photos and videos; build galleries; manage admission announcements, downloads, vacancies, leadership profiles, contacts, homepage banners, menus and FAQs; update social links; control SEO metadata; schedule publication; and archive outdated content.

CMS menu: Dashboard, Pages, News, Events, Admissions, Gallery, Videos, Downloads, Vacancies, Staff Profiles, Homepage, Menus, Forms, SEO, Social Media, Media Library, Pending Approval, Published, Archived, Analytics, Settings.

C8. CMS content workflow

No user should be able to create and publish immediately by default.

1. Content author writes a **draft**.
2. Editor **reviews** and edits.
3. Content approver **approves** or **returns** it for correction.
4. Approved content is **scheduled or published** to the public site.

C9. CMS roles

Role	Can
Content Author	Create drafts
Content Editor	Edit and review
Media Manager	Manage photos and videos
CMS Publisher	Publish approved content
CMS Administrator	Configure the CMS
Head Teacher / authorised approver	Approve sensitive institutional publications

Technical and institutional authority stay separate: ICT maintains the CMS, Communications prepares news, Administration reviews institutional information, and the Head Teacher or an authorised officer approves important publications.

C10. SEO requirements

Unique titles, meta descriptions, a clean H1/H2 hierarchy, readable URLs, canonical URLs on the real domain (see A2 item 1), alt text, internal links, XML sitemap, robots.txt, structured data and Open Graph tags. Target searches: *Alpha Adventist Pre & Primary School, Adventist school Kigoma, primary school Kigoma, Christian school Kigoma, ICT school Kigoma, primary school admissions Kigoma* — and their Kiswahili equivalents.

Part D — Staff portal, HR, documents and notices

The Alpha Staff Portal is the employee's digital workplace: each person sees their own records and tasks, and extra modules appear only with extra permissions.

D1. Staff portal menu and dashboard

Menu: Dashboard, My Profile, Leave, My Contract, My Documents, Staff Documents, Notice Board, Downloads, Calendar, Training, My Requests, Notifications, Help, Logout.

The dashboard greets the employee by name and shows four tiles — leave balance (for example 18 days), contract status, new notices and new documents — above upcoming events such as staff meetings, training and school events.

D2. Staff profile

Fields: Staff ID, full name, position, department, employment type, employment date, supervisor, official email, official telephone, contract status. HR-only information stays restricted.

D3. Leave management

Leave types: annual, sick, maternity, paternity, compassionate, study, emergency, other authorised leave.

Leave form: employee name and Staff ID (auto-filled), department, leave type, start date, return date, number of days, reason, handover employee, emergency contact, attachment, comments → *Submit for approval*.

Approval route (configurable per leave type):

1. Employee submits the application.
2. Immediate supervisor reviews.
3. Head of Department reviews.
4. HR / Administration reviews.
5. Head Teacher or authorised approver decides; the employee is notified of approval or rejection.

Not every leave type needs all stages; Administration sets the routes in the workflow engine (E2).

Statuses: Draft, Submitted, Supervisor Review, HR Review, Final Approval, Approved, Returned for Correction, Rejected, Cancelled.

D4. Contract management

Under **My Contract**, each employee securely sees only their own employment contract, appointment letter, renewals, job description, promotion letters, confidentiality agreement, policy acknowledgements and training agreements. No ordinary employee can see another's contract.

Contract workflow:

1. HR drafts the contract.
2. Administration reviews.
3. Finance reviews, when financial terms need checking.
4. Authorised management approves.
5. Head Teacher or authorised signatory signs.
6. Employee acknowledges and signs.
7. Contract becomes active and is stored in the secure archive.

Expiry reminders at 90, 60, 30, 14 and 7 days, sent to HR, the employee, the supervisor and Administration.

D5. Staff Document Library

Category	Examples
HR	HR policy, staff handbook, leave policy, employment forms
Academic	Curriculum guidance, lesson plan templates, assessment guidelines, examination procedures
Administration	Internal circulars, SOPs, internal forms
ICT	ICT policy, acceptable use policy, cybersecurity and password guidance
Child protection	Safeguarding policy, child protection and reporting procedures
Safety	Emergency procedures, health and safety

D6. Document classification

Level	Class	Who can see it
1	Public	Anyone
2	Internal	Authorised employees
3	Restricted	Specific departments or roles
4	Confidential	Named authorised users or roles
5	Highly confidential	Strict HR, executive or legal access

Metadata on every controlled document: document ID, title, category, department, owner, classification, version, status, effective date, review date, expiry date, created by, reviewed by, approved by, approval date. Example: ALPHA-HR-POL-001, version 2.0, Internal, owned by Human Resources.

Lifecycle: Draft → Under Review → Changes Requested → Revised → Approved → Published/Issued → Review Due → Superseded → Archived. An old controlled document is superseded and archived, never simply deleted.

D7. Digital staff notice board

- **Categories:** general, academic, HR, administration, ICT, meetings, training, events, emergency, deadlines.
- **Priorities:** Urgent, Important, Information, General.
- **Each notice holds:** title, message, category, audience, priority, publisher, publication date, expiry date, attachment, and whether acknowledgement is required.
- **Targeting:** all staff, or specific groups — teachers, administration, HR, finance, ICT, security, drivers, kitchen, boarding staff, heads of department, or named individuals — to avoid information overload.
- **Read and acknowledge:** important notices carry an "I have read and understood" button. The system records Staff ID, notice ID and version, read date, acknowledgement date and session, and shows management the share of staff who have acknowledged and who is outstanding.

Part E — Permissions, workflows and oversight

Access is decided by role-based access control plus fine-grained, action-based permissions — never a simple admin/user split — and every approval runs through one configurable workflow engine that records each decision.

E1. Roles and permissions

Core roles: Super/System Administrator, Head Teacher, Deputy Head Teacher, School Administrator, HR Officer, Finance Officer, Academic Coordinator, Head of Department, ICT Officer, DPO/Privacy Officer, CMS Manager, Content Editor, Teacher, Supporting Staff, Auditor, and later Parent and Student.

Permissions are actions, not roles:

staff.view_self	staff.view_department	staff.view_all	
leave.create	leave.review	leave.approve	
contract.view_self	contract.create	contract.review	contract.approve
document.create	document.read	document.review	document.approve
document.publish	document.archive		
cms.create	cms.edit	cms.review	cms.publish
notice.create	notice.publish		
audit.read			

Example permission matrix:

Resource	Staff	HOD	HR	Admin	Head Teacher	ICT
Own contract	Yes	Yes	Yes	Restricted	Yes	No
Other contracts	No	No	Yes	Restricted	Yes	No
Own leave	Yes	Yes	Yes	Yes	Yes	No
Department leave	No	Yes	Yes	Yes	Yes	No
Payroll	No	No	Restricted	Restricted	Yes	No
Staff handbook	Yes	Yes	Yes	Yes	Yes	Yes
HR investigation	No	No	Restricted	Restricted	Restricted	No
Website CMS	No	No	Limited	Yes	Oversight	Yes
Server settings	No	No	No	No	No	Yes

System administrator ≠ business approver. ICT may manage servers, accounts, database availability, backups and integrations, but that never lets ICT approve leave, read salary records, approve contracts, inspect disciplinary files or change HR decisions.

E2. Workflow and approval engine

Approval logic is configured once, not hard-coded per module. Core objects: WorkflowDefinition, WorkflowStep, Request, Requester, Approver, Decision, Comment, Attachment, Escalation, Notification, AuditEvent.

Example definitions:

```
Workflow: LEAVE_REQUEST
  Step 1 = Supervisor
  Step 2 = HR
  Step 3 = Head Teacher
  On approve: update leave status, notify employee
```

```
Workflow: EMPLOYMENT_CONTRACT
  Step 1 = HR
  Step 2 = Administration
  Step 3 = Finance
  Step 4 = Authorised signatory
  Step 5 = Employee acknowledgement
```

Document routes:

Document	Route
HR policy	HR author → HR manager → Administration → Head Teacher
ICT policy	ICT officer → DPO / security review → Administration → Head Teacher
Finance document	Finance officer → senior finance / administration → authorised management
Academic document	Teacher → Head of Department → Academic Coordinator → Head Teacher

Approval matrix:

Process	Level 1	Level 2	Level 3	Final
Leave	Supervisor	HOD	HR / Admin	Authorised management
Contract	HR	Admin	Finance, if applicable	Authorised signatory
HR policy	HR	Admin	—	Head Teacher
Academic policy	HOD	Academic Coordinator	Admin	Head Teacher
ICT policy	ICT	DPO / Security	Admin	Management
Website news	Editor	Communications	Admin	Publisher
Child media	Media officer	Consent check	Admin	Authorised publisher
Financial form	Finance	Admin	—	Authorised approver
Public circular	Author	Administration	—	Authorised signatory

Approver actions: Approve, Reject, Return for correction, Comment, Delegate (where policy allows), Escalate. Every action is recorded.

E3. Dashboards

Dashboard	Shows
Administration	Staff: total, on leave, new, contract. Approvals: leave, contracts, documents pending. Compliance: policies due for review, expired documents, missing acknowledgements. Website: draft news, pending publications, traffic. System: active users, security alerts, recent activity
HR	Total, permanent and contract staff; staff on leave; pending leave; contracts expiring; missing staff documents; pending contract approvals; training due; policies awaiting acknowledgement

E4. Search, notifications and calendar

- **Search** across documents, notices, policies, forms, news, events, and — where authorised — staff and contracts. Filters: department, category, year, status, classification, owner, version. Results always obey the user's permissions.
- **Notification Centre:** in-app, email, optional SMS and future push. Examples: "Your leave request LEV-2026-034 has been approved", "Contract CTR-2026-087 expires in 30 days", "A new staff notice requires acknowledgement", "Document ALPHA-HR-POL-004 requires your approval".
- **School calendar:** opening, holidays, examinations, staff and parent meetings, sports, worship, training, graduation, school events and deadlines, each marked Public, Staff only or Selected users.

E5. Audit trail

Every sensitive operation writes a strongly protected audit record: date and time, user, action, resource, result and session/IP — for example, HR_MANAGER_01 approved contract CTR-2026-087 on 4 October 2026 at 09:43, success.

Audited events: login, failed login, logout, download, upload, edit, approval, rejection, permission change, role assignment, document access, account creation and archival.

E6. Reporting

Area	Reports
HR	Total staff, leave statistics, contract expiry, employment categories
Documents	Pending approval, due for review, expired, superseded
CMS	Published news, pending content, website traffic
Security	Failed logins, permission changes, unauthorised access attempts, security alerts

Part F — Admissions, portals, privacy and child protection

Every service that touches children's or parents' data is built privacy-first: consent is recorded, access is limited to the right family, and sensitive records never appear publicly.

F1. Online admissions

1. Parent starts an application.
2. Enters student information, then guardian information.
3. Chooses programme and class (KG I – Standard VII; day or boarding).
4. Uploads required documents and submits.
5. Admissions office reviews.
6. Assessment or interview where applicable.
7. Decision and admission offer.
8. Enrolment.

Statuses: Draft, Submitted, Documents Required, Under Review, Assessment, Accepted, Waitlisted, Declined, Enrolled.

Until this is built, the current enquiry forms must at least alert the office and give parents a reference number (A2 item 4).

F2. Parent portal (future)

Child profile, results, attendance, fees, receipts, homework, timetable, school notices, calendar, teacher communication, documents and consent management. A parent sees only the children they are authorised to represent.

F3. Student portal (future)

A child-friendly interface for homework, timetable, learning materials, announcements, results where appropriate, the school calendar and ICT learning resources. It grows out of today's Kids Zone, keeping its rules: no external links, ads, chat or data collection from children.

F4. Child media consent register

For each student, the parent or guardian grants or refuses consent separately for: website, Facebook, Instagram, YouTube, printed materials and external promotion. Each record holds who granted it, the date, the categories, status, review or expiry date, and any withdrawal. Photos are removed within 48 hours when consent is withdrawn, and children are never named beside their photos.

F5. Privacy components

- Website, staff, and parent/student privacy notices
- Cookie controls where applicable
- Data protection policy and data retention schedule
- Media consent register (F4)
- Data subject request procedure
- Incident and breach management
- Access control, audit logging, secure deletion and backup controls
- A named safeguarding contact and a way to report concerns (Law of the Child Act, 2009)

F6. Never public

Student IDs, birth certificates, medical records, home addresses, detailed attendance, academic reports, private parent records, passwords, disciplinary records, staff contracts, salary records, HR investigations and any confidential document.

F7. Tanzania Personal Data Protection Act alignment

The school should confirm its duties as a data controller under the Personal Data Protection Act, 2022, including any registration with the Personal Data Protection Commission, and sign processor agreements with any technology company that handles personal data on its behalf. Privacy must shape the design from the start: database design → permissions → APIs → storage → logging → backups → retention → deletion. This document is not legal advice; confirm obligations with a Tanzanian lawyer or the Commission.

Part G — Security, data, APIs, hosting and mobile

The server — never the browser — decides who can see what: every private request is authenticated, permission-checked against the specific resource, and logged, and default access is denied.

G1. Minimum security requirements

HTTPS; multi-factor authentication for privileged users; strong authentication and secure password hashing; rate limiting (per user, not only per spoofable IP — see A2 item 6); session timeout; secure cookies; CSRF, XSS and SQL-injection protection; secure file uploads with malware and type validation; server-side authorisation; encryption; backup security; API security; security headers; dependency management; vulnerability testing; audit logging.

G2. Server-side authorisation

Every private API request passes four checks, in order; any "no" returns 403 Forbidden:

1. Is the user authenticated?
2. Do they hold the needed permission?
3. May they access **this specific** resource (ownership, department, classification)?
4. Is the resource in a valid state for this action?

Hiding a button is not security.

G3. Secure document download

Staff portal → authenticated user → document request → authorisation engine → classification check → ownership/role check → audit event → **temporary, expiring download link**. Staff contracts and confidential records never get permanent public URLs.

G4. Data model

Area	Tables
Access	users, roles, permissions, user_roles, role_permissions
Staff	departments, staff_profiles
Leave	leave_types, leave_requests, leave_balances
Contracts	contracts, contract_versions
Documents	documents, document_versions, document_categories, document_permissions
Workflow	workflow_definitions, workflow_steps, workflow_instances, approvals
Notices	notices, notice_audiences, notice_acknowledgements
Website	pages, news, events, media, menus, downloads
Students	students, guardians, guardian_students, media_consent
Admissions	admission_applications
System	notifications, audit_logs, system_settings

Key entities:

Document:	id, document_number, title, description, category_id, classification,
	department_id, owner_id, current_version, status, effective_date, review_date, expiry_date, created_by, created_at, updated_at
DocumentVersion:	id, document_id, version_number, storage_key, checksum, uploaded_by, change_summary, approval_status, created_at
Approval:	id, workflow_instance_id, step_id, request_id, approver_id,
decision,	comment, decision_at
AuditLog:	id, user_id, action, resource_type, resource_id, ip_address,
user_agent,	timestamp, result, metadata

The current site stores everything in one JSON file (data/db.json); moving to a proper database is part of Phase 1.

G5. API structure

```
/api/auth /api/users /api/roles /api/permissions
/api/staff /api/departments
/api/leave /api/contracts
/api/documents /api/downloads
/api/workflows /api/approvals
/api/notices /api/calendar
/api/cms/pages /api/cms/news /api/cms/events /api/cms/media
/api/admissions
/api/notifications /api/audit
```

G6. Hosting and environments

Move off the development-style setup to: official domain → DNS → CDN / web application firewall → application layer → frontend and API → database → secure storage → backup.

Keep three separate environments: **Development** (programmers), **Staging** (testing and user acceptance) and **Production** (the live platform). Production is never used for development or testing.

G7. Backups

Asset	Backup
Database	Automated daily
Documents	Versioned
Code	Source-control repository (private)

Asset	Backup
Media	Replicated storage
Configuration	Secured copy

Restores are tested on a schedule; an untested backup is not a reliable backup.

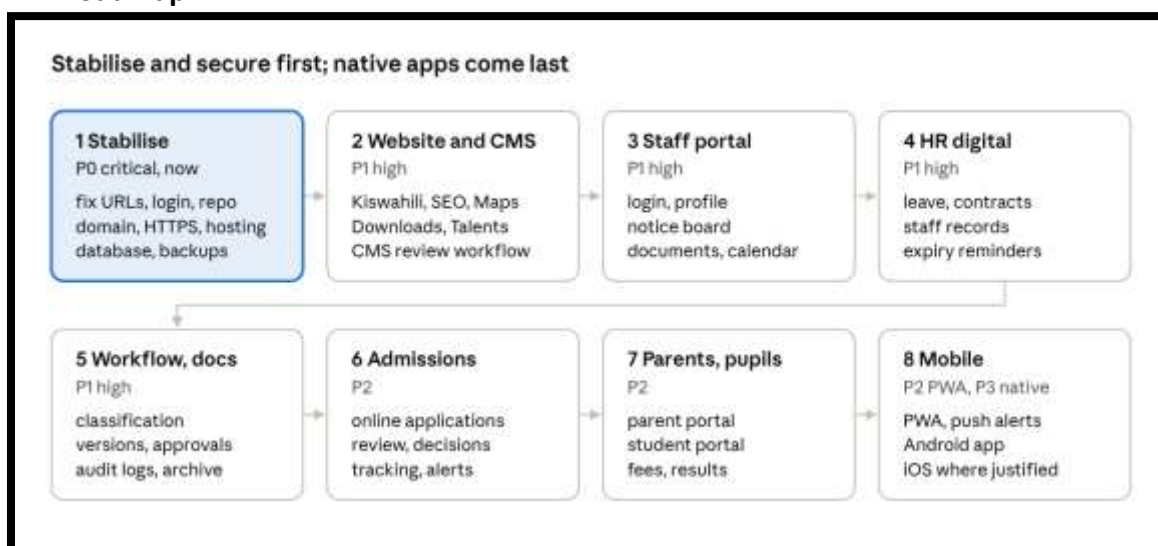
G8. Mobile

- **Mobile-first is mandatory:** desktop, laptop, tablet, Android and iPhone; staff can apply for leave or approve a document from an authorised phone.
- **PWA first:** install to phone, staff login, notices, leave, documents, calendar, notifications and limited offline information.
- **Alpha School App later,** Android first, with three experiences: *Parent* (fees, attendance, results, homework, notices, messages), *Staff* (leave, contracts, documents, notices, approvals) and *Management* (dashboards, approvals, alerts, reports).
- Until the portals exist, use WhatsApp Business and bulk SMS for parent notices, since many parents use basic phones.

Part H — Roadmap, priorities, rules and conclusion

Delivery runs in eight phases; Phase 1 absorbs every critical audit finding, so no new module is built on an unsafe or invisible site.

H1. Roadmap



Delivery roadmap · 8 phases in order, not to scale

Phase 1 covers audit items A2.1–A2.6 plus production hosting, the official domain, HTTPS, a real database, backups and monitoring; Phase 2 carries the visibility work from A3–A4.

H2. Priority matrix

Component	Priority
Fix dead-domain canonical/share URLs, robots.txt, default admin password	P0 critical
Make repository private; purge child photos and internal files	P0 critical
Resolve 503 / production hosting	P0 critical
Security architecture and authentication	P0 critical
Official domain and email	P0 critical
Admissions alerts on existing forms	P0 critical
Public website, Kiswahili version, SEO, Google Business Profile	P1 high
CMS and Download Centre	P1 high

Staff portal and notice board	P1 high
Leave and contract management	P1 high
Document management, permission engine, workflow and approvals	P1 high
Audit trail, privacy and consent	P1 high
Online admissions	P2
Parent and student portals	P2
PWA	P2
Native mobile app	P3

H3. Programmer non-negotiable rules

1. Authentication does not mean authorisation.
2. Every private request passes server-side permission checks.
3. Access is denied by default unless explicitly granted.
4. Staff access only authorised records.
5. An employee's contract is never visible to other ordinary employees.
6. Technical administrators do not automatically receive HR business authority.
7. CMS authoring and publishing are separate permissions.
8. Sensitive approvals are auditable.
9. Controlled documents are versioned.
10. Old approved documents are superseded or archived, never silently overwritten.
11. Private documents never use unrestricted public URLs.
12. Child-related data receives enhanced protection.
13. Media publication follows authorisation and consent rules.
14. Production, staging and development are separate.
15. Privileged users use MFA.
16. Backups are automatic and restoration is tested.
17. Mobile responsiveness is mandatory.
18. Workflow rules are configurable.
19. Security events are logged.
20. Public and confidential storage are logically separated.
21. No credentials, personal data or child photos are ever committed to a code repository.

H4. Conclusion

The strongest outcome is not a website with many pages but a secure, integrated digital school where each person gets the services their role needs: parents discover, trust and apply; staff work and track requests; managers approve and monitor; HR manages people and compliance; the CMS team publishes through review; ICT operates and recovers; and leadership governs. The current site is a good foundation — once Phase 1 makes it safe and findable, every later phase builds on it.

Sources

- [Alpha Adventist website repository \(GitHub\)](#) — source reviewed for Part A
- [Alpha Adventist Primary School — Encyclopedia of Seventh-day Adventists](#)
- Alpha Digital School Platform blueprint (attached requirements document) — basis for Parts B–H